



Utilizing the NDWI and CCM with Heatmap Analysis Waterbody Fluctuations in Al-Hawizeh Transboundary Marshes: A Strategy for Supporting Decision-Makers Planning in Achieving Sustainability

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Abstract: Wetlands are fundamental habitats for migratory birds and species habiting shallow waters. In this study, we quantitatively analyze the surface water area of fluctuations in the Al-Hawizeh Marshes, a transboundary wetland shared by Iraq and Iran. Following a severe drought in the past decade, these marshes have shown ecological recovery, positioning them today as a sustainable ecosystem. The study examines whether these marshes are once again facing the risk of drought or will continue along a trajectory of ecological conservation. This study employs Landsat satellite imagery spanning nearly a decade to monitor hydrologic dynamics for the 2015, 2018, 2021 and 2024 calendar years by relying on the computational capabilities of Google Earth Engine (GEE) platform. In parallel, the normalised difference water index (NDWI) was applied to delineate water bodies and quantify the spatial extent of surface water. The year 2021 proved to be the most anomalous in terms of water area, presenting an average of 448.4 km², in sharp contrast with the severe desiccation monitored over the years, including 2018 (48.4 km²) and 2024 (49.6 km²). The results demonstrate the utility of remote sensing for monitoring these largely inaccessible wetlands and provide vital, data-driven evidence of the critically endangered status of Al-Hawizeh Marshes. This article attains particular importance not only through its spatial analysis and statistical evaluation, employing the correlation coefficient matrix (CCM) and heatmap analysis effectively illustrating the fluctuations revealed across monthly and annual classifications, but also through the results it presents, which indicate that the shallow water bodies are undergoing a gradual recession and are generally progressing toward desiccation. Accordingly, the findings call for rapid solutions in the form of watershed-based transboundary water management agreements, along with a deeper exploration of the drivers behind such extreme hydrological regime shifts, in order to support decision-maker planning in conserving this ecologically rich corner of the world. This approach aims to ensure the continuity of the ecological environment, safeguard the local community, and ultimately achieve sustainability.

Keywords: Biodiversity conservation; Climate change impacts; Google Earth Engine; Heatmap analysis; Spatial extent of surface water; Local community preservation; Sustainability; Transboundary wetlands management

1 Introduction

Wetland ecosystems are among the most biologically productive and diverse habitats on our planet, serving as hubs for numerous ecosystem services that benefit both the environment and human societies. The most important of these services are water filtration, flood regulation, bank stabilization, carbon storage, the provision of habitats for a vast array of plant and animal species [1, 2]. Wetlands are estimated to cover only 6% of the total land area, yet they are the primary ecological home of over 40% of the global species underscoring their ecological significance [3]. Despite their crucial role, Wetlands are among the most fragile ecosystems on Earth. Since the early 20th century, it is estimated that approximately 70% of wetlands worldwide have disappeared [4, 5]. The persistence and protection of these critical ecosystems depend on the success of water management, which requires the timely collection of accurate and reliable information regarding the spatial extent and temporal dynamics of water distribution for wetland planning.

Because of the extensive, unique, and inaccessible nature of wetlands, the conventional field monitoring methods are often impractical for extensive areas. In this context, remote sensing technologies have emerged as indispensable tools [6–8].

Satellite remote sensing is considered a cost-effective, environmentally friendly, and advanced method for monitoring soil water saturation over extended periods [9]. Given the importance of spectral indices derived from multispectral imagery in accurately mapping and monitoring water bodies and their variations, these indices provide essential information to water resource managers and decision-makers [10–13].

Accordingly, the study employed these indices to analyze the marshland regions of southern Iraq, with particular focus on the Al-Hawizeh Marshes.

In the 1990s, a large-scale engineering project was implemented to drain the marshes, which reduced their area by more than 90%. However, the collapse of these ecosystems was not solely a natural phenomenon. This was described as one of the most severe human-made environmental disasters of the twentieth century [14–18].

Similarly, the importance of the Al-Hawizeh Marshes lies in their designation as a Ramsar site and their inclusion on the United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage List. This significance stems not only from their role as a habitat for numerous ecological species inhabiting shallow waters, but also from their remarkable resilience to recover after 2003, marked by the return of water to the marshland areas and the restoration of ecological diversity [6, 17, 19].

So, the aim of this article is to monitor the fragile environment of the Al-Hawizeh Marshes in order to determine whether they are undergoing an ecological period of recovery, which could restore their capacity to provide ecosystem services and sustain biodiversity, or whether they are experiencing hydrological instability leading to recession and desiccation, placing them in a state of ecological decline. Accordingly, this paper seeks to deliver a rigorous scientific assessment that contributes to understanding the future of this wetland and underscores the urgent need for water management strategies to support decision-making and achieve both environmental and cultural sustainability.

2 Methodology

The analysis was conducted within the GEE environment Figure 1.

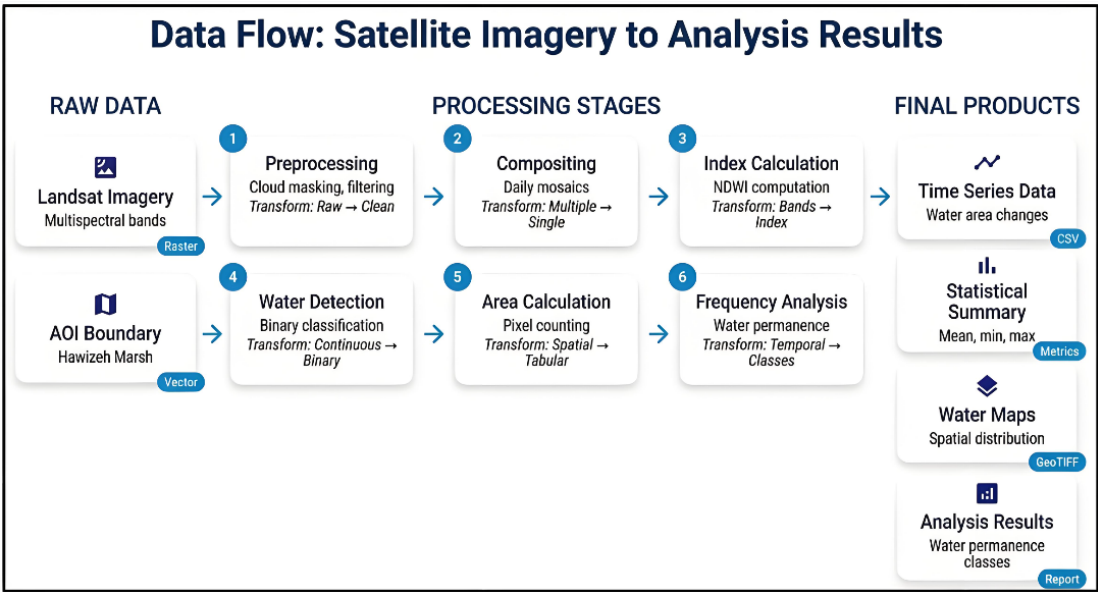


Figure 1. Methodology process to the study

- Step 1. **Study Area:** The official vector border was imported into GEE and defined as the Area of Interest (AOI).
- Step 2. **Satellite Data Preparation:** Landsat 5, 7, and 8 Surface Reflectance collections were filtered for the years (2015, 2018, 2021 and 2024) and constrained to AOI.
- Step 3. **Composite Images:** To mitigate data gaps (e.g., Landsat 7 SLC-off) and ensure consistency a daily composite (mosaic) image was generated.
- Step 4. **Water Detection:** Normalised difference water index (NDWI) were calculated for each daily composite image, and the results were used to create a binary mask (0 is land, 1 is water) to classify each pixel. The threshold of 0.2 is applied, which helps to distinguish water bodies and shallow waters, reducing errors caused by reflectance from vegetation or moist soil. The (QA_RADSAT) mask was used to exclude saturated data (e.g., highly bright areas such as sand or strong water reflections [19, 20]).

Step 5. **Calculate Water Area:** Using the binary water masks, the total water area within the AOI was calculated as a percentage.

Step 6. **Data Export:** Time-series data of water area statistics were exported as a CSV files; derived raster maps were exported as GeoTIFF files.

Step 7. **Analysis:** Summary statistics (mean, minimum, maximum) were calculated for the water area and coverage metrics. Water permanence was analyzed seasonally and annually to provide deeper insights into the hydrological stability of the Marshes.

3 Methods and Materials

3.1 Study Area

The focus is on Al-Hawizeh Marshes, a vast, transboundary wetland ecosystem situated along the border between southeastern Iraq and southwestern Iran [21]. Within Iraq, they extend approximately from 47°25'E to 47°45' E, and from 31°05'N to 31°40' N [22]. It varies in size between summer and winter and from one year to another, this environment supports the local communities whose livelihoods (the Marsh inhabitants) depend on its natural resources [10]. Its hydrological system exhibits considerable complexity, as it is fed mainly by branches of the Tigris River in Iraq and the Karkheh River in Iran. The official boundaries of the protected area, whose total area amounts to 1,377 km² Figure 2.

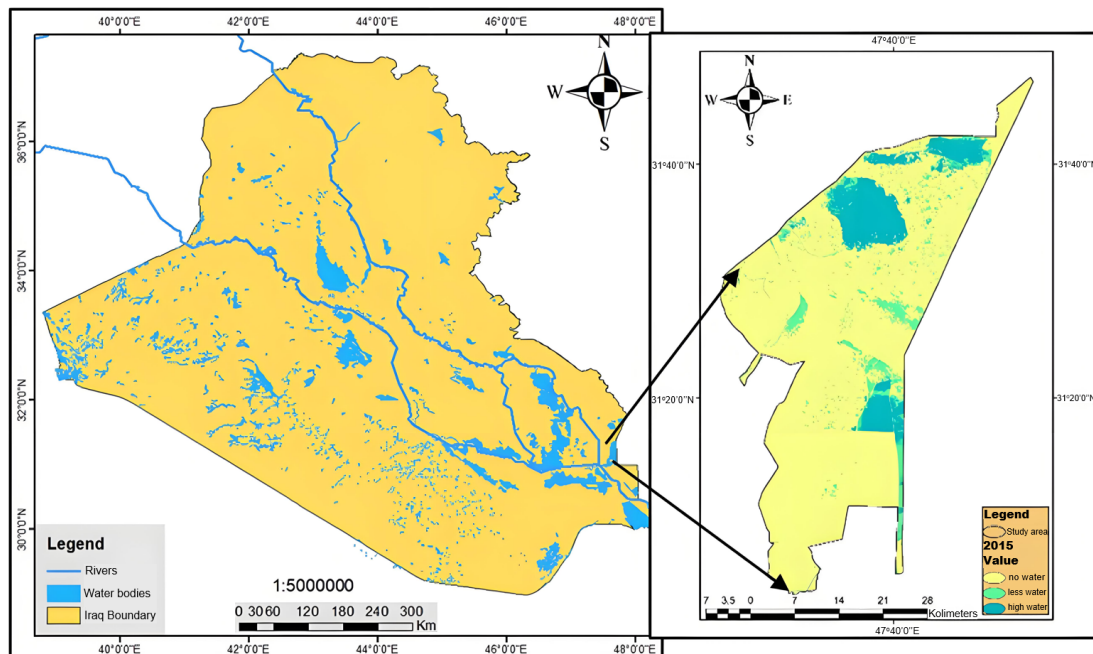


Figure 2. Study area of the transboundary Al-Hawizeh Marshes in Iraq

Source: This figure was prepared by the authors depended on ArcMap 10.8 and Google Earth Engine.

This delineation represents the legally designated extent, which does not correspond to the actual water coverage area because it changes between season and from year to year. That boundary defines of the protected AOI. The official boundaries of the protected area cover a total area of 1,377 km².

3.2 Data Acquisition and Pre-Processing

The primary data source for this research was the collection of Landsat satellite imagery, specifically Tier one Surface reflectance products from Landsat 5, Landsat 7, and Landsat 8 in Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+), and Operational Land Imager (OLI), respectively. These collections provide atmospherically corrected data [23]. The analysis was conducted for the entirety of the years 2015, 2018, 2021, and 2024 thus enhancing the accuracy of the following water detection analysis [24, 25].

3.3 Water Body Extraction Using Normalised Difference Water Index

The core of the water detection methodology was NDWI, proposed by Xu [26]. The spectral index is based on the difference in reflectance between water in the green and near-infrared (NIR) regions of the electromagnetic spectrum.

Water bodies absorb NIR radiation significantly, while exhibiting strong reflection in the green band [26]. The index is calculated using the following formula:

$$\text{MNDWI} = \frac{(\text{Green} - \text{SWIR1})}{(\text{Green} + \text{SWIR1})} \quad (1)$$

For Landsat 8, the NDWI is calculated using (Band 3) and (Band 5), and for otherwise, it is (Band 2–Band 4) [27].

Although there is a variety of advanced indices such as the Modified NDWI (MNDWI) [28]; this index was adopted for computational efficiency, and demonstrated performance in delineating open water bodies across vast vegetated wetland landscapes such as Al-Hawizeh Marshes [16].

4 Results and Discussion

The analysis yielded detailed quantitative and visual results, revealing significant and dramatic fluctuations in the waterbody area of Al-Hawizeh Marshes across the selected years. The findings are presented in a comparative data table that summarizing the mean monthly water area.

4.1 Comparative Analysis of Mean Water Area (2015–2024)

This result reveals a significant variation in the Marshes hydrological condition from year to year, as seen in Table 1.

Table 1. Comparative mean monthly water area (km²) for years 2015, 2018, 2021, and 2024

Month	Mean Water Area in 2015 (km ²)	Mean Water Area in 2018 (km ²)	Mean Water Area in 2021 (km ²)	Mean Water Area in 2024 (km ²)
January	64.22	90.98	538.06	5.96
February	N/A	88.56	574.32	28.7
March	99.21	95.94	629.1	38.37
April	163.46	73.78	521.27	138.79
May	77.03	46.14	533.23	97.40
June	116.67	75.43	23.23	92.18
July	89.6	71.72	518.76	45.79
August	40.31	17.48	443.58	15.18
September	66.32	10.89	385.68	30.16
October	43.22	9.82	215.01	38.54
November	91.53	0.03	266.22	27.87
December	63.92	0.00	232.01	35.77
Average	83.23	48.4	448.4	49.56

In contrast, years 2018 and 2024 exhibit severely desiccated conditions, with annual average water areas of only 48.4 km² and 49.6 km², respectively. The year 2015 represents a more moderate, baseline condition, with an average water area of 83.2 km². With an average annual water area of 448.4 km², year 2021 stands out as an extraordinarily wet year. This extreme fluctuation between extensive inundation and severe dryness underscores the dynamic and vulnerable nature of the marsh's ecosystem.

To substantiate this, each year will be analyzed separately using GEE maps and statistical analysis as listed in table separately.

4.2 Year Analysis

4.2.1 Analysis for the year 2015

In 2015, Al-Hawizeh Marshes exhibited moderate water levels with distinct seasonal variation, representing a baseline condition for this study. The annual average water area was approximately 83.2 km². A time-series chart for 2015 (Figure 3) and the water distribution map as shown in Figure 4, confirm a typical hydrological pattern for the region.

Figure 3 illustrates how the average water area changed during 2015. It exhibits a clear peak in spring (April) at 163.5 km², followed by a decrease to its lowest point in late summer/early autumn (September) at 40.3 km², before rebounding towards recovery at the end of the year.

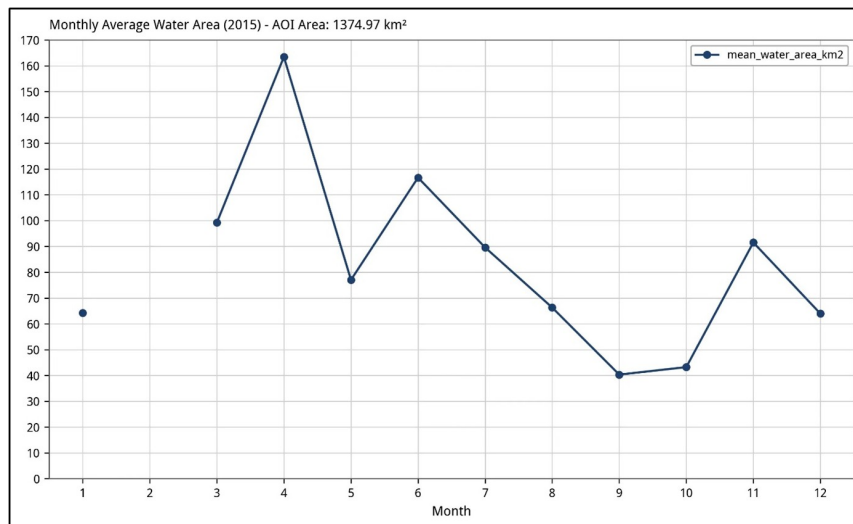


Figure 3. Monthly average water area (2015)

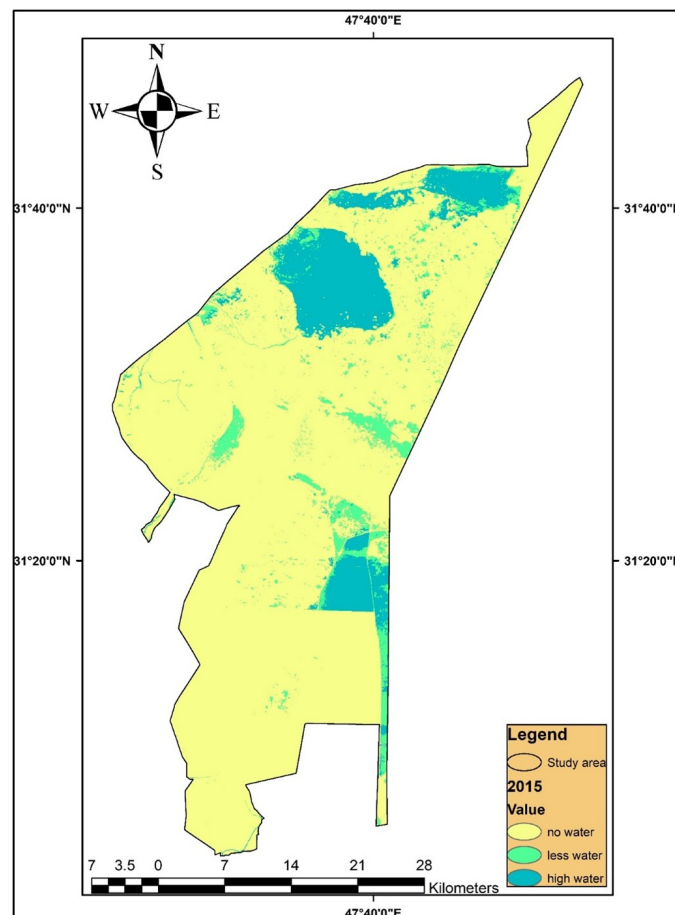


Figure 4. Water distribution in Al-Hawizeh Marshes (2015)

Source: This figure was prepared by the authors depended on ArcMap 10.8 and Google Earth Engine.

4.2.2 Analysis for the year 2018

The year 2018 marked a period of significant dryness, with the annual average water area plummeting to 48.4 km². This is illustrated in the monthly average chart (Figure 5) and the spatial distribution map for 2018 (Figure 6).

This chart Figure 5 shows a peak water area in March (95.9 km²), followed by a decline that accelerates dramatically in the second half of the year. By the end of the year, the water area falls to a lower value.

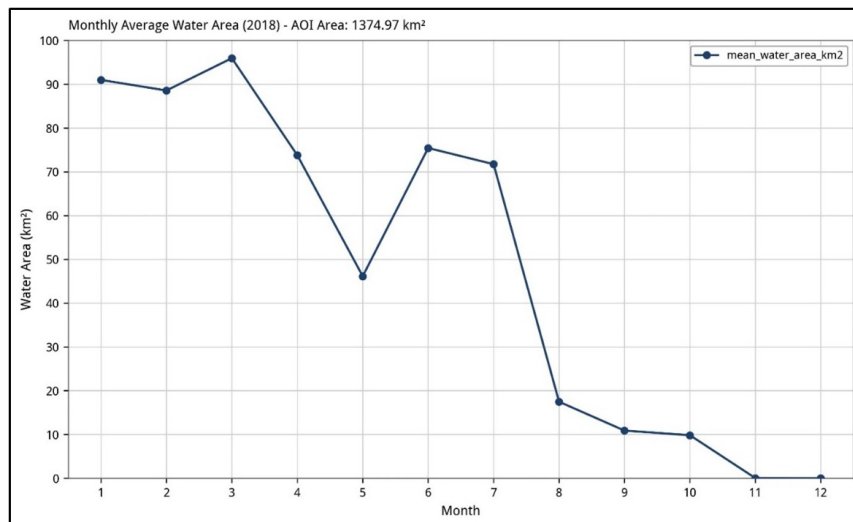


Figure 5. Monthly average water area (2018)

Source: This figure was prepared by the authors depended on ArcMap 10.8 and Google Earth Engine.

The map Figure 6 for 2018 illustrates a substantial reduction in water coverage compared to 2015. Large expanses are classified as “no water”, dominated by yellow, with only small, fragmented pockets of “high water” visible, highlighting a critical dry period.

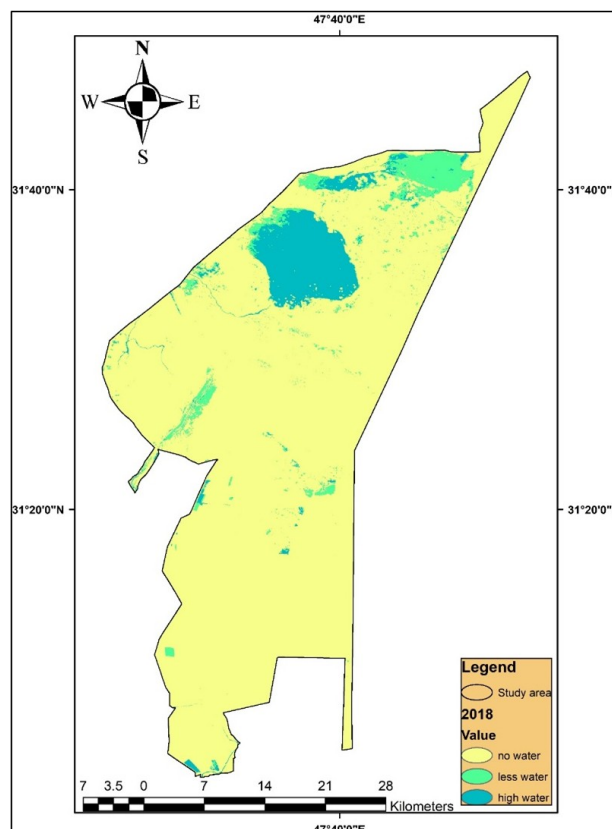


Figure 6. Water distribution in Al-Hawizeh Marshes (2018)

Source: This figure was prepared by the authors depended on ArcMap 10.8 and Google Earth Engine.

4.2.3 Analysis for the year 2021

In stark contrast to the rest of study period, 2021 was an exceptionally wet year. The average water area surged 448.4 km², around ten times the 2018 average. This is clearly illustrated in the monthly average chart Figure 7, and the water distribution map Figure 8.

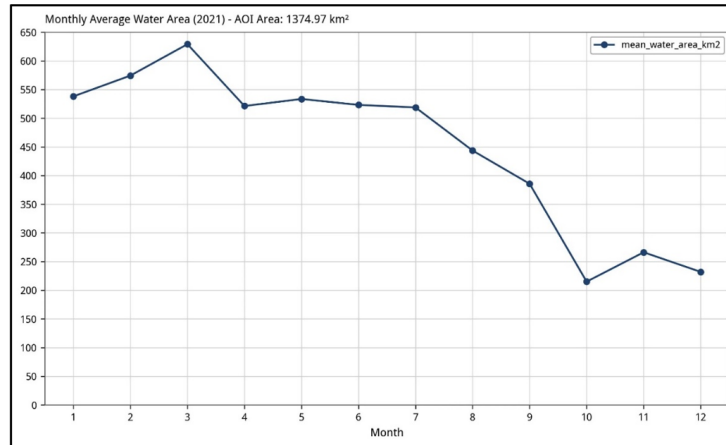


Figure 7. Monthly average water area (2021)

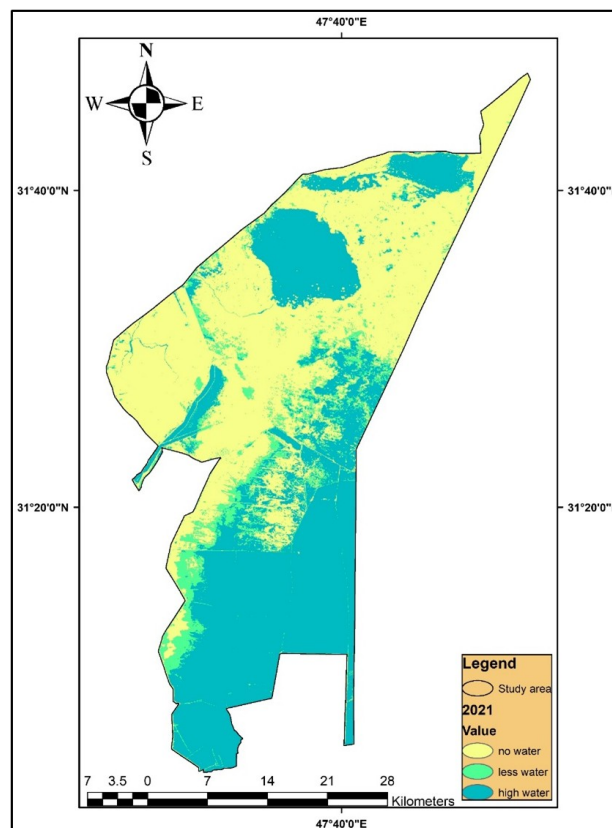


Figure 8. Water distribution in Al-Hawizeh Marshes (2021)

Source: This figure was prepared by the authors depended on ArcMap 10.8 and Google Earth Engine.

Figure 7 represents a dramatic reversal from the conditions observed in 2018. The monthly chart Figure 7 shows consistently high-water values throughout the year, peaking in March at 629.1 km² and remaining above 200 km² even at its lowest point in October. This suggests a major flood event or a period of sustained, unusually high-water supply to the marsh system. Furthermore, the chart for 2021 depicts water area on a completely different scale from the other years. Levels remain exceptionally high for the entire year, peaking in March and gradually decreasing, but never approaching the low levels recorded in other years. It should be noted that the chart in Figure 7 differs from charts in Figures 3, 5, and 9, where the y axis values do not exceed 170 km² and are divided into intervals of 10 km². But, the y axis in the year 2021 rises to 650 km², with the lowest values reaching 200 km², and the divisions set at 50 km². This distinction reflects the markedly higher water volumes observed in 2021, which is characterized as a flood year with generally elevated water levels.

This map Figure 8 illustrates the extensive flooding in 2021. Most of the AOI is covered by water, areas of “high

water” (blue), representing a dramatic reversal from the conditions seen in 2018.

The interpretation of 2021 reveals the emergence of large volumes of water, as shown by statistical and spatial analysis. This is attributed to the fact that it followed the year of the COVID-19 pandemic, which allowed the environment to recover. One of the most prominent reasons was the increase in rainfall and the inflow of water from upstream sources. It also reflects the policy of ineffective water management across Iraq.

4.2.4 Analysis for the year 2024

In 2024, the marsh conditions appeared to revert to a dry state, with hydrological patterns closely resembling those of 2018. The annual average water area was 49.6 km², almost identical to the 2018 average. The time-series chart Figure 9 and map Figure 10 show that fluctuation.

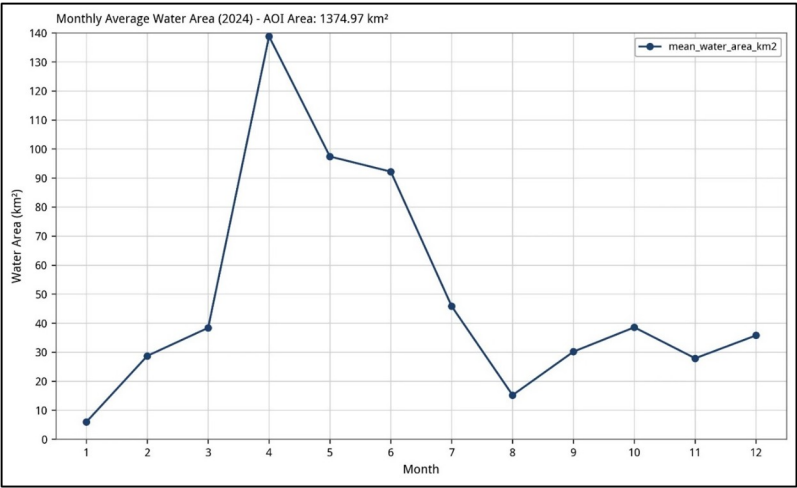


Figure 9. Monthly average water area (2024)

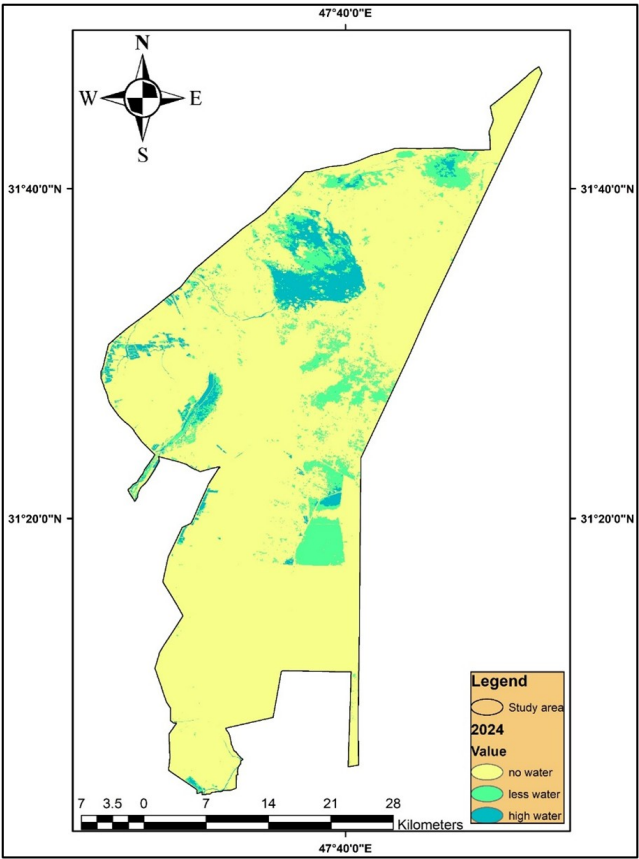


Figure 10. Water distribution in Al-Hawizeh Marshes (2024)

The time-series chart Figure 9 reveals a notable spring peak in April (138.8 km²), which was surprisingly higher than the peaks in 2015 or 2018. However, this wet period was short-lived and was followed by a rapid decline to very low levels in late summer (15.2 km² in August).

The 2024 map Figure 10 shows a landscape largely devoid of surface water, closely resembling the 2018 map. It is dominated once again by “no water” (yellow) areas, with only sparse and fragmented water presence in blue. The overall condition in 2024 reflects a return to significant water scarcity after the extreme wet period of 2021.

Findings from the Al-Hawizeh Marshes between 2015 and 2024 reveal a system under extreme stress, characterized by profound inter-annual volatility with significant implications for its ecological functions and the human communities it supports. These results are summarized in Figure 11. As well, the results were further visualized through a heatmap, which highlights the intensity and density of waterbody values using color gradients, effectively illustrating the fluctuations revealed by the study (Figure 12).

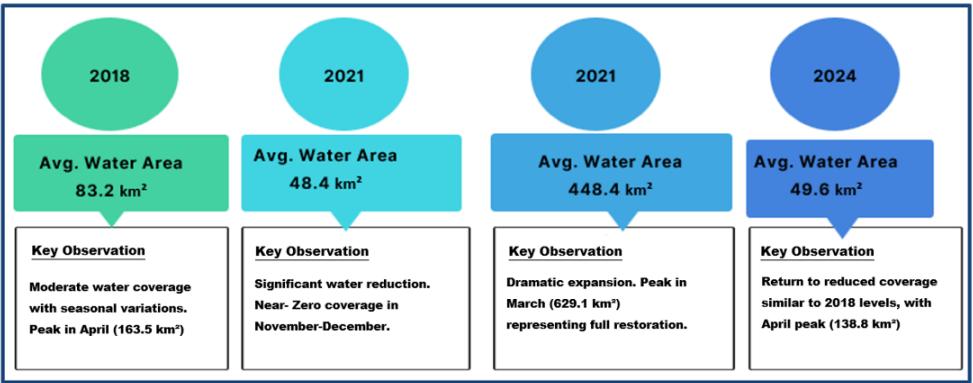


Figure 11. Infographic demonstration of study area results

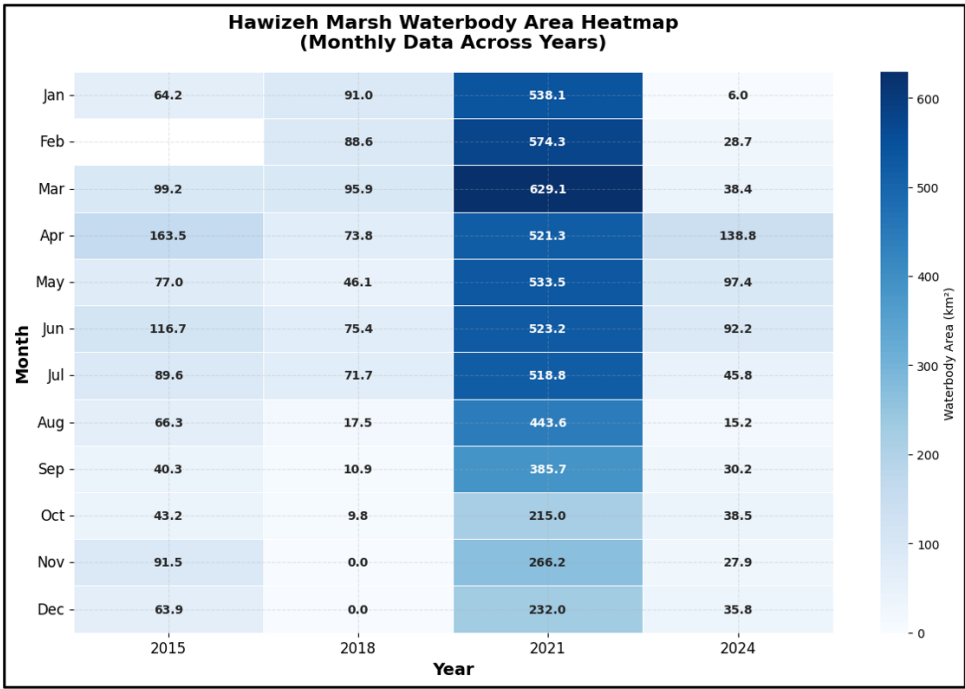


Figure 12. Heatmap plot for Al-Hawizeh Marshes waterbody area during the study period

The heatmap in Figure 12 illustrates that the dark blue color, representing higher waterbody values, reached approximately 600 km² for about seven months during 2021. In contrast, it declined to lighter shades of blue, reaching the palest level during the driest months of 2018 and 2024. The correlation coefficient matrix (CCM), displayed in Table 2, which along with the accompanying heatmap, clarifies the linear correlations between the monthly waterbody area dynamics of Al-Hawizeh Marshes across the observed years (2015, 2018, 2021, 2024).

Table 2. The correlation matrix is computed using the Pearson correlation coefficient

Year	2015	2018	2021	2024
2015	1	0.621	-0.425	0.501
2018	0.621	1	-0.796	0.642
2021	-0.425	-0.796	1	-0.812
2024	0.501	0.642	-0.812	1

The interpretation in Table 2 of results based on Pairwise Correlation Analysis is as follows:

1. Years 2015 and 2018 ($r = 0.621$): A moderate positive correlation indicates shared seasonal hydrological patterns, with both years exhibiting peak water areas during spring (March–April) and declines in autumn (October–December). However, the consistently lower water areas in 2018 (e.g., near-zero values in November–December) suggest intensified drought conditions relative to 2015, despite the structural similarity in seasonal cycles.

2. Year 2021 and other years ($r = -0.425$ to -0.812): In 2021, there are moderate negative correlations with 2015 ($r = -0.425$), and high negative correlations with 2018 ($r = -0.796$) and 2024 ($r = -0.812$).

3. This inverse relationship is attributable to the exceptional hydrological conditions in 2021, where mean water areas exceeded 500 km² during peak months (e.g., 629 km² in March). In contrast, 2018 and 2024 represent severe drought years (mean areas < 100 km²), while 2015 reflects intermediate conditions. The pronounced negative correlations underscore 2021 as an outlier driven by anomalous flooding, disrupting typical interannual coherence.

4. Years 2018 and 2024 ($r = 0.642$): A robust positive correlation highlights the shared drought-driven hydrological regimes of these years. Both exhibit minimal water retention during winter months (e.g., 0 km² in December 2018; 5.96 km² in January 2024) and limited recovery in summer. This alignment suggests recurrent drought stressors, though 2024 shows marginally higher resilience (e.g., 138.8 km² in April 2024 vs. 73.8 km² in April 2018).

5. Years 2015 and 2024 ($r = 0.501$): The moderate positive correlation reflects analogous seasonal phasing, with peak areas occurring in spring (April–June). However, the significantly reduced water extents in 2024 (e.g., 97.4 km² in May 2024 vs. 77.0 km² in May 2015) indicate progressive aridification, potentially linked to long-term climate stressors.

This strengthens the notion that, through spatial analysis and statistical assessment grounded in extracted indicators, it demonstrates that the Al-Hawizeh Marshes are in recession and are generally moving toward desiccation.

5 Conclusions

The methodology of this study, which relies on the GEE platform and Landsat satellite data, has proven highly effective in monitoring the Al-Hawizeh Marshes ecosystem during the period between 2015 and 2024. This large and complex system is characterized by extreme hydrological fluctuations, with the marshes experiencing severe drought in 2018 and 2024, contrasted by extensive flooding in 2021.

The results underscore the critical state of the marshes and the urgent need for cooperative transboundary water management. Sustainable and equitable water sharing that ensures environmental flow requirements is essential for the marshes' long-term existence, their distinctive biodiversity, and the preservation of the cultural heritage of ancient civilization.

Preserving their sustainability necessitates a comprehensive approach that takes into account crucial factors:

Firstly, preserving environmental flow: ensuring sufficient water inflows throughout the year to sustain the marshes distinctive ecology. Secondly, regional cooperation: recognizing the transboundary nature of these essential resources. Finally, community engagement: supporting residents' livelihoods and empowering them to take part in decision-making and resource management.

To ensure the ability of this fragile ecological fractal system to endure in the long term, adaptive planning that adopts resilient and sustainable strategies to deal with drought and floods together represents a rational methodology to water management, which contribute to supporting the decision-maker planning and to achievement of sustainability.

Author Contributions

Conceptualization, S.R.A.A.-T. and A.S.D.; methodology, A.S.D.; software, S.R.A.A.-T. and A.S.D.; validation, S.R.A.A.-T.; formal analysis, S.R.A.A.-T. and A.S.D.; investigation, S.R.A.A.-T. and A.S.D.; resources, S.R.A.A.-T. and A.S.D.; data curation, S.R.A.A.-T.; writing—original draft preparation, A.S.D.; writing—review and editing, S.R.A.A.-T. and A.S.D.; visualization, S.R.A.A.-T. and A.S.D.; supervision, S.R.A.A.-T. and A.S.D.; project administration, S.R.A.A.-T. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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